The Relationship of Permeability and Permittivity at the Perfect Matching Point of Electromagnetic Wave Absorption for the Absorber Filled by Metallic Magnetic Particles

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The electromagnetic wave absorption properties of flake-shaped carbonyl-iron/paraffin composites with various volume concentrations were calculated by the measured complex permeability and permittivity. The complete electromagnetic wave absorption was searched for the volume concentration of 0.125, 0.15 and 0.20 at 5.8, 4, and 3.6 GHz, respectively. The permittivity is much larger than the permeability at the complete absorption frequency points. Based on the origin of the absorbing peaks, the contour map displaying the relationship between the complex permittivity and complex permeability at the perfect matching point was developed. The complex permittivity obtained from the contour map basically agrees with the experimental data for a given complex permeability. Hence, the contour map can be used to design the absorber for obtaining the excellent electromagnetic wave absorption.

Index Terms—**EM materials, composite materials, matching properties, reflection properties, permeability, permittivity.**

I. Introduction

With the development of electronic communication technology, electromagnetic pollution has attracted considerable attention. To solve this problem, electromagnetic-wave-absorbing material with high performance is needed [1]–[3]. For an absorber, its complex permeability and permittivity play key roles in determining the electromagnetic wave absorption properties. According to the transmission line theory, when a wave is normally incident to an absorber layer terminated by a metal plate, the reflection loss (RL) at a given absorber thickness is calculated using the following equation [4], [5]:

$$\mathrm{RL} = -20\log\left|\frac{Z_{\mathrm{in}}-1}{Z_{\mathrm{in}}+1}\right| \tag{1}$$

where Z_{in} is the normalized input impedance of the absorber. The Z_{in} is described by

$$Z_{\mathrm{in}} = \sqrt{\mu_r/\varepsilon_r}\tanh\left[j(2\pi ft/c)\sqrt{\mu_r\varepsilon_r}\right] \tag{2}$$

where f is the frequency, t is the thickness of the absorber, c is the velocity of light, and μ_r and ε_r are the complex permeability and permittivity of the composite, respectively. Equation (2) indicates that Z_{in} is closely related to μ_r, ε_r, f and t. To obtain the complete absorption, which corresponds to a negative infinite RL value in (1), Z_{in} should be equal to 1. The approximately complete absorption of electromagnetic wave has been reported in many absorbers, and in [7]–[10], it was proposed that the complex permeability needs to be equal to the complex permittivity for obtaining the good absorption performance. It can be seen from (2), if the thickness of absorber is infinite, the complete absorption ($Z_{\mathrm{in}} = 1$) of electromagnetic wave can be achieved when $\mu_r = \varepsilon_r$.

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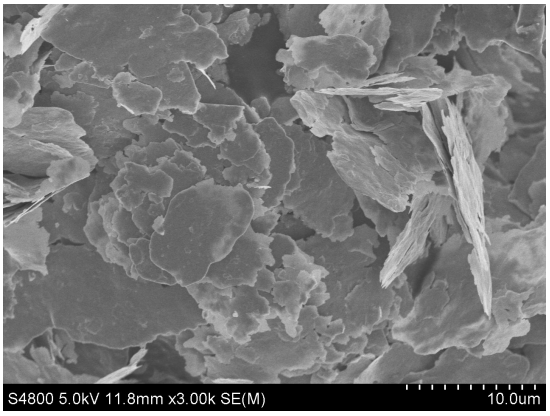


Fig. 1. Scanning electron microscope image of ball-milled carbonyl-iron particles.

In practical application, the thickness of absorber is impossible to be infinite and to be as thin as possible. The composites filled with metallic magnetic particles can be made into thin absorber due to their high permeability and permittivity [11], [12], and have continuously attracted much attention in recent years. For this kind of absorber, ε_r is generally much larger than μ_r. Hence, the electromagnetic loss inside the absorber is finite, because only a part of electromagnetic wave can enter the absorber from the air–absorber interface. However, the complete absorption of electromagnetic wave for the absorber filled with metallic magnetic particles can be achieved by the quarter-wavelength cancellation [12]–[16]. In the reported literatures in which strong absorption peaks were achieved, most of the studies paid attention only to the achievement of the strong absorption peak, and a few studies dealt with the relationship between complex permeability and complex permittivity at the perfect absorption frequency point. As we know, the absorption performance of an absorber is determined by its

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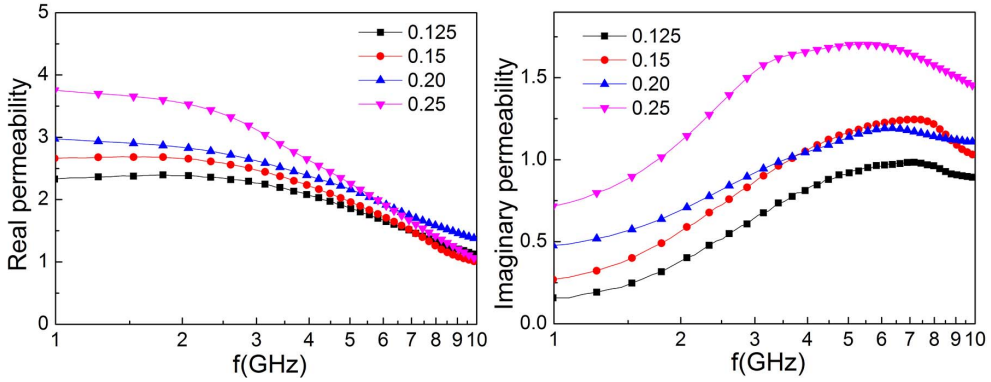


Fig. 2. Frequency dependence of the complex permeability for the flake-shaped carbonyl-iron powder/paraffin composites with different volume fractions.

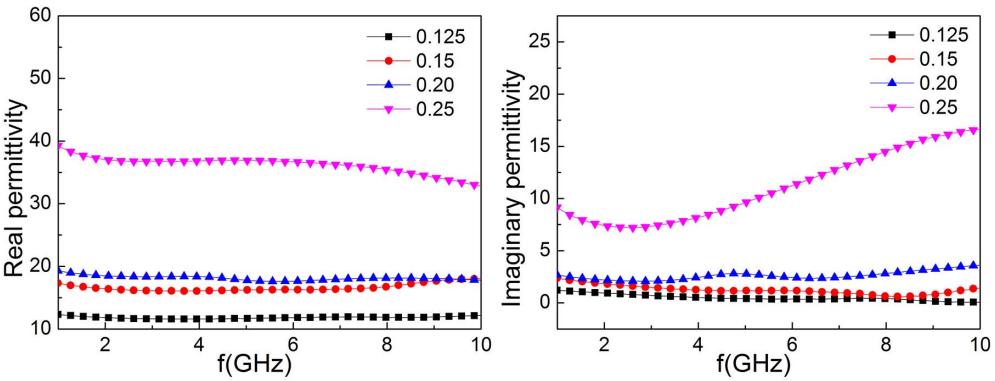


Fig. 3. Frequency dependence of the complex permittivity for the flake-shaped carbonyl-iron powder/paraffin composites with different volume fractions.

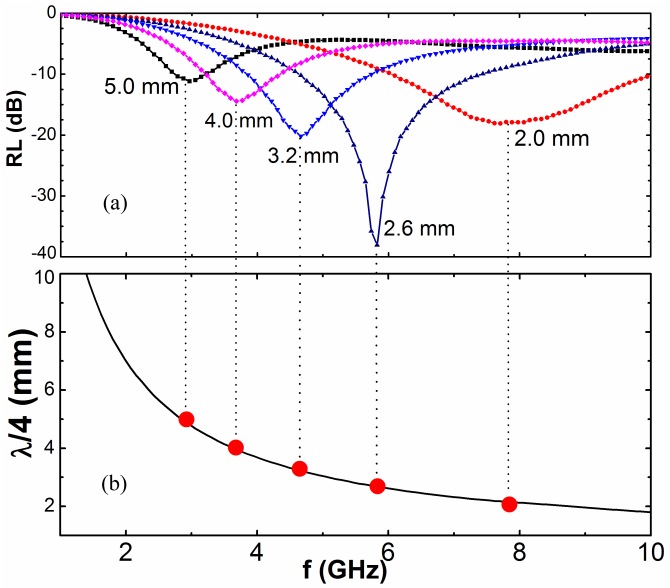


Fig. 4. (a) Frequency dependence of RL for flake-shaped carbonyl-iron particle/paraffin composites with $V_c = 0.125$ at various thicknesses. (b) Frequency dependence of $\lambda/4$ of the composite. Red dots in (b): corresponding absorber thicknesses at the RL peak frequencies in (a).

complex permeability and complex permittivity. The deep investigation on their relationship at the perfect matching point is vital to design the high-performance absorber by adjusting the electromagnetic parameters. In this paper, the widely reported flake-shaped carbonyl-iron powder was selected as metallic magnetic fillers due to its high permeability in gigahertz frequency range [13], [14]. The electromagnetic wave absorption properties of flake-shaped carbonyl-iron/paraffin composites with various volume concentrations are calculated using (1) and (2). The perfect matching points are found for the composites with $V_c = 0.125$, 0.15, and 0.20, and their electromagnetic parameters at the perfect matching frequency are listed. On the other hand, the contour map displaying the relationship between complex permittivity and complex permeability at the perfect match point of an absorber was developed based on the origin of absorption peaks. The relationship of μ_r and ε_r from the experimental data and that from the contour map are compared.

II. Experiments

The flake-shaped carbonyl-iron powder was prepared using a planetary ball mill in alcohol. Both the ball and the bowl were made of hardened steel. The morphology of the milled particles was characterized using a scanning electron microscope (SEM). The flake-shaped carbonyl-iron/paraffin composites were prepared by uniformly mixing the magnetic particles in paraffin. The volume concentration of flake-shaped carbonyl-iron is 0.125, 0.15, 0.20, and 0.25. After mixing, the mixture was pressed into a toroidal shape with an outer diameter of 7 mm, an inner diameter of 3.04 mm, and a thickness of ~1 mm. The toroidal samples were measured using the Agilent E8363B vector network analyzer in the frequency range of 1–10 GHz. The complex permeability

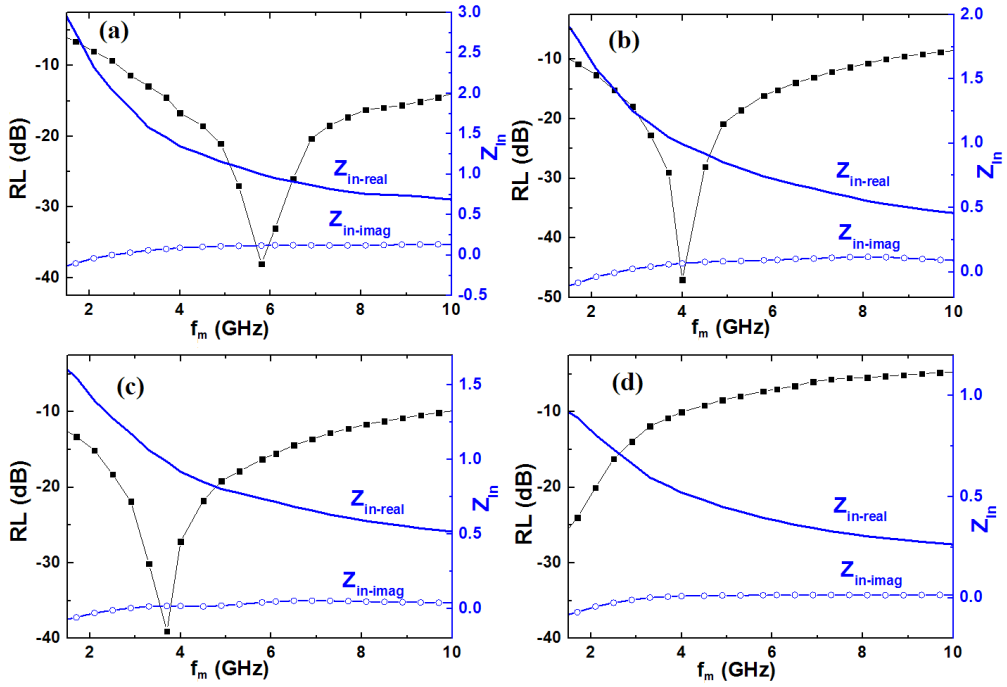


Fig. 5. Matching frequency dependence of RL peak value and normalized input impedance for composites with (a) $V_c = 0.125$, (b) 0.15, (c) 0.20, and (d) 0.25.

TABLE I

PARAMETERS OF FLAKE-SHAPED CARBONYL-IRON PARTICLE/PARAFFIN COMPOSITES AT THE PERFECT MATCHING POINT

V_C	RL (dB)	f_{pm} (GHz)	t_{pm}(mm)	ε	ε (Fig. 6)	μ
0.125	-38	5.8	2.6	11.8-0.4 j	7.1-1.5 j	1.7-1.0 j
0.15	-47	4.0	2.8	16.1-1.2 j	11.3-2.0 j	2.2-1.1 j
0.20	-40	3.6	3.0	18.4-2.2 j	15.6-2.7 j	2.5-1.0 j

and permittivity were evaluated by measuring the reflection coefficient S_{11} and the transmission coefficient S_{21} in the APC7 coaxial mode.

III. RESULTS AND DISCUSSION

For identifying the morphology of ball-milled carbonyl-iron particles, the SEM micrograph was analyzed. Fig. 1 shows the SEM image of ball-milled carbonyl-iron particles. A series of flake-shaped carbonyl-iron particles was obtained after milling. The diameter of the particles is ~10 um, the thickness is ~0.2 um, and the aspect ratio is ~50:1. The large aspect ratio results in an easy magnetization plane. In materials with easy-plane anisotropy, high permeability and resonance frequency can be obtained simultaneously [17].

The frequency dependence of complex permeability and complex permittivity for the flake-shaped carbonyl-iron/paraffin composites with various concentrations are shown in Figs. 2 and 3, respectively. It is observed that both complex permeability and permittivity increase with the increasing V_c, and the complex permittivity is much larger than the complex permeability at a given V_c in the whole measurement frequency range. When an electromagnetic wave is normally incident to a material layer, the reflection coefficient at the air–material interface is calculated using the following equation:

$$\Gamma = \frac{\sqrt{\mu_r/\varepsilon_r} - 1}{\sqrt{\mu_r/\varepsilon_r} + 1}. \tag{3}$$

Considering that the permittivity is much larger than the permeability for a given composite, a part of electromagnetic wave must be reflected from the air–absorber interface. This means the complete entrance of electromagnetic wave into the absorber is impossible for the above composites. To further investigate the electromagnetic wave absorption properties of the composites, the RL properties of the composites are calculated using (1) and (2). Fig. 4(a) shows the dependence of RL on frequency at various absorber thicknesses for the composite with $V_c = 0.125$. For a given absorber thickness, a RL peak appears at a frequency. The RL property is sensitive to the absorber thickness and the RL peak shifts to lower frequency with the increasing thickness. This phenomenon has been found in other absorbers filled with metallic magnetic particles and successfully explained by the quarter-wavelength cancellation at the air–absorber interface [15], [18], [19]. The quarter-wavelength cancellation can be described as follows. When an electromagnetic wave is normally incident to an absorber backed by a metal plate, it is partially reflected at the air–absorber interface of the absorber. There is another

emerging wave at this interface that is reflected completely at the absorber–metal plate interface and then emerges from the air–absorber interface. At the special frequency point, the two reflected waves cancel each other at the air–absorber interface when they are out of phase by 180°, and hence results in a RL peak [18]. The calculated quarter wavelength in the composite with frequency using its complex permeability and permittivity is shown in Fig. 4(b), as a black line. The absorber thicknesses corresponding to the RL peak frequencies in Fig. 4(a) are indicated as red dots in Fig. 4(b) at their corresponding frequencies. It is clearly seen that the absorber thickness agrees well with the quarter-wavelength values of the composite at any given frequency. The above results confirm that the RL peaks are attributed to the quarter-wavelength cancellation.

From Fig. 4(a), it is known that there is a RL peak value at a frequency point in the RL curve for a given absorber thickness t. The corresponding absorber thickness and RL peak frequency are defined as matching thickness t_m and matching frequency f_m, respectively. The perfect matching thickness $t_{\rm pm}$ and perfect matching frequency $f_{\rm pm}$ occur when the complete absorption that corresponds to the negative infinite RL value occurs. If the RL peak values at every matching frequency are recorded, the optimal absorption performance (RL peak value) dependence of matching frequency can be displayed. The results are shown in Fig. 5(a). Besides the RL peak value, the normalized input impendence $Z_{\rm in}$, including the real part $Z_{\rm in-real}$ and imaginary part $Z_{\rm in-imag}$, at the RL peak value for every matching frequency is also listed in this figure. The dependence of the RL peak values and $Z_{\rm in}$ on matching frequency for the other composites with $V_c = 0.15$, 0.20, and 0.25 are shown in Fig. 5(b)–(d), respectively. For $V_c = 0.125$, 0.15, and 0.20, the minimum RL peak values are -38, -47, and -40 dB, respectively. When $V_c = 0.25$, the minimum RL peak value is only -25 dB. The electromagnetic wave absorption percentage is 99.9% at -30 dB, so the RL value smaller than -30 dB can be considered as complete absorption. It is clearly seen that $Z_{\rm in}$ is very close to 1 when the composites achieve the complete electromagnetic wave absorption. The RL value, $f_{\rm pm}$, $t_{\rm pm}$, ε_r, and μ_r at $f_{\rm pm}$ are listed in Table I. As can be seen from Table I, the ε_r and μ_r values at the perfect matching point are not equal and the value of ε_r is apparently higher than μ_r. With the increase of μ_r, the ε_r increases correspondingly. The experimental results show that the complex permeability does not need to be equal to complex permittivity when the complete absorption is achieved at a small thickness for the absorber filled with metallic magnetic particles.

In order to investigate the relationship between the complex permittivity and the complex permeability at the perfect matching point, the formation mechanism of the RL peak must be used, since the origin of the RL peak comes from quarter-wavelength cancellation. The relationship between RL peak frequency and absorber thickness can be described by [16], [18]

$$t_m = \frac{nc}{4 f_m \sqrt{|\mu_r \varepsilon_r|}} \qquad (n = 1, 3, 5, \ldots,) \tag{4}$$

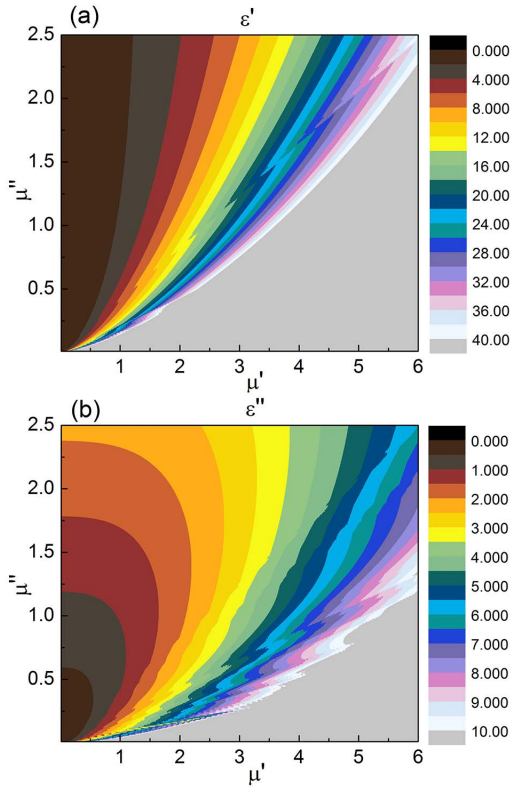


Fig. 6. Contour map of complex permittivity varied with complex permeability at the perfect matching point. (a) Real part of permittivity. (b) Imaginary part of permittivity.

where t_m is the matching thickness and f_m is the matching frequency. As shown in (4), t_m is inversely proportional to the f_m, and the corresponding phenomena have been shown in Fig. 4(b). Considering that the thin absorber is preferable, only $n = 1$ is studied in this paper. If the input impendence only at the matching point is considered, (2) can be transformed into

$$Z_{\rm in} = \sqrt{\mu_r/\varepsilon_r}\tanh(j(2\pi f_m t_m/c)\sqrt{\mu_r \varepsilon_r}). \tag{5}$$

By substituting the matching thickness in (5) with (4), the normalized input impendence of (5) at the matching point can be described by

$$Z_{\rm in} = \sqrt{\frac{\mu_r}{\varepsilon_r}}\tanh\left(j\frac{\pi}{2}\frac{\sqrt{\mu_r \varepsilon_r}}{\sqrt{|\mu_r \varepsilon_r|}}\right). \tag{6}$$

In this case, $Z_{\rm in}$ is determined only by μ_r and ε_r. At the perfect matching point, $Z_{\rm in}$ should be equal to 1. If $Z_{\rm in}$ is fixed at 1, the relationship between μ_r and ε_r at the perfect matching point can be obtained by numerical calculation. Fig. 6(a) and (b) shows the calculated contour maps for ε' and ε'' with the variation of μ' and μ'' at the perfect matching point.

From Fig. 6, to achieve complete electromagnetic wave absorption, a complex permittivity can be obtained based on the known complex permeability. In Table I, the complex permeability at the perfect matching frequency for the composites with $Vc = 0.125$, 0.15, and 0.20 has been listed. Based on

these complex permeability values, the corresponding complex permittivity values for $Z_{\text{in}} = 1$ can be read from Fig. 6, and the results are shown in Table I. We can see that the calculated permittivity is also much larger than the permeability and basically agrees with the measured permittivity. The large ratio of ε_r/μ_r for the complete absorption can be understood by the quarter-wavelength model. In the quarter-wavelength cancellation, the RL peak value is determined by the energy difference of the two waves reflected from the air–absorber interface and the absorber–metal plate interface. $Z_{\text{in}} = 1$ (RL $= -\infty$) corresponds to the same energy of the two reflected waves [16]. To ensure that the two waves cancel each other completely, a part of electromagnetic wave must be reflected from the air–absorber interface, so a certain ratio of ε_r to μ_r is necessary. In Table I, we can see that the measured complex permittivity is somewhat larger than the calculated value. The reason is explained as follows. When the contour map is developed, the Z_{in} is strictly fixed at 1. However, the perfect matching points we search are not the complete absorption strictly, but correspond to the finite RL values. This may result in the slight difference of the complex permittivity from the calculation and measurement.

IV. Conclusion

Flake-shaped carbonyl-iron particle/paraffin composites are found to have the perfect matching point at $V_c = 0.125$, 0.15, and 0.20. At the perfect matching point, the complex permittivity is much larger than the complex permeability. Based on the origin of the absorption peak, the relationship of permittivity and permeability at the perfect matching point is developed in the contour map. The relationship of complex permeability and complex permittivity from the contour map and experimental data are basic in agreement. The contour map gives us a straight way for judging whether the complete electromagnetic wave absorption exists or improving the electromagnetic wave absorption performance by adjusting the permittivity. Since the contour map is developed from quarter-wavelength cancellation, the map is suitable for the absorbers in which a complete absorption is generated by quarter-wavelength cancellation.

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